

Lab 7/8 – MATH 240 – Computational Statistics

Reagan Sernick
rsernick@colgate.edu

Abstract

This paper explores the beta distribution, and methods of estimating parameters α and β from a data set to create a realistic probability model.

Keywords: **Beta Distribution:** Distribution of a random variable from 0-1 that takes parameters α and β ; **Estimators:** Estimations of distribution parameters based on a sample; **Sampling Distributions:** A sample of random samples that help find population parameters more accurately

1 Introduction

The beta distribution is a powerful and flexible probability model, especially useful for modeling random variables constrained between 0 and 1, like proportions or probabilities. In this lab, we dive deep into the mechanics of the beta distribution, exploring how its shape parameters, α and β , influence the distribution's behavior. Through visualization, simulation, and statistical estimation, we analyze how beta distributions can model real-world data, with a particular focus on using Method of Moments (MOM) and Maximum Likelihood Estimation (MLE) to fit a model to actual death rate data. By comparing these estimation techniques, we gain insight into their accuracy and reliability, building a clearer picture of how well theory translates to practice.

2 Methods

2.1 How do Parameters Change a Distribution?

For the first part of this lab we created four sample beta distributions with different α and β values using `dbeta()`.

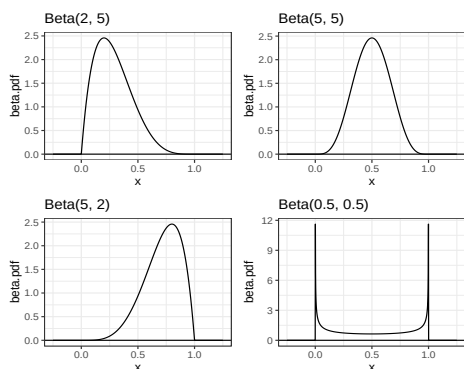


Figure 1: Sample Beta Distributions

These graphs show that as you change the shape parameters of a distribution the mean, variance, skewness, and kurtosis change.

What Else Can Shape Parameters Tell Us?

In order to see how the shape parameters affected mean, variance, skewness and kurtosis we calculated them from the formulas using α and β and the more precise moment calculation.

The comparison in Table 1 shows that using α and β to calculate mean, variance, skewness and kurtosis is accurate on average up to 4 decimal places.

2.2 How Well Does a Beta Distribution fit a Random Distribution?

Next, to see how a beta distribution could fit a real random distribution we used `rbeta()` to create random distributions of size $n = 500$ using the same shape parameters as the distributions in the previous part.

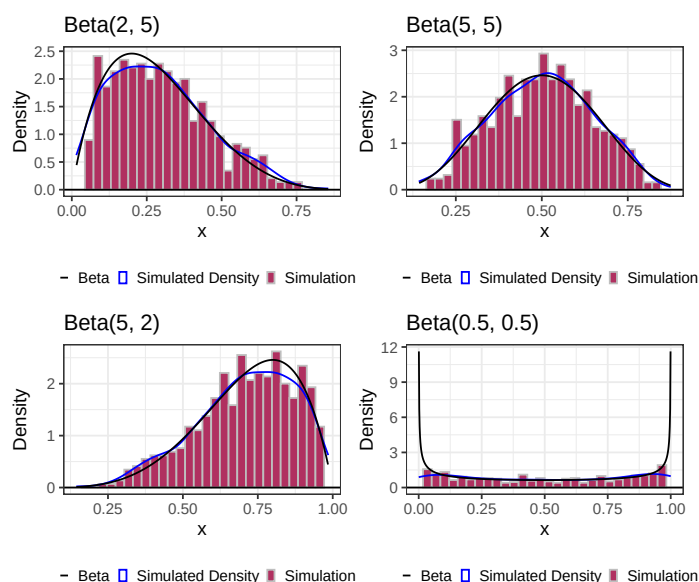


Figure 2: Simulated Beta Distributions

As seen in the graphs, the simulation's density matches the sample beta curve fairly well, showing that

2.3 Why does Sample Size Matter?

Sample size is important when trying to estimate a population parameter because the larger the sample size the more your sample distribution will match the population. In order to visualize this, we used the `cumstats` (Erdely and Castillo, 2017) package and created graphs showing how mean, variance, skewness, and kurtosis change as sample size increases.

The `cumulative stats` graph shows that as sample size increases, the sample statistic gets closer to the true parameter.

2.4 Multiple Samples? (*Sampling Distribution*)

In order to see how taking multiple samples can help estimate the population parameters, we ran 1,000 simulations of random beta samples where $n = 500$, $\alpha = 2$, and $\beta = 5$.

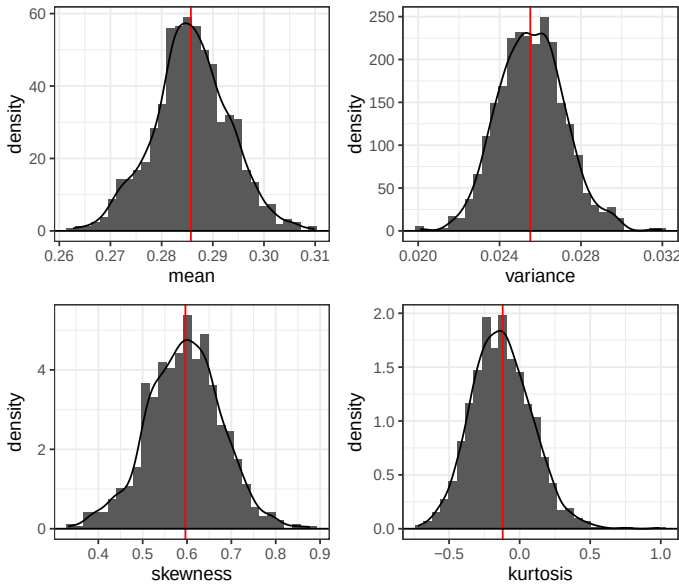


Figure 3: Sampling Distributions

With the red vertical lines representing the population parameter, we can see that the mean of statistics of the 1,000 samples represent the population parameter fairly accurately

3 Estimators

Estimators are useful to help us fit a model to a real random variable based on data. The two main methods to get estimators are Method of Moments (MOM) and Maximum Likelihood Estimation (MLE).

The data we used to fit a distribution was of death rates in different countries around the world (World Bank, 2024). Using both MLE and MOM estimates for α and β we modeled world death rates using a beta distribution.

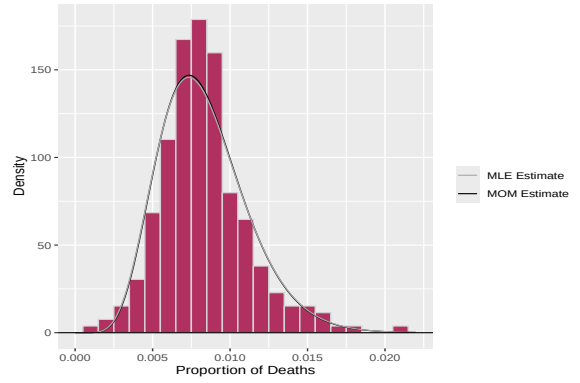


Figure 4: Death Data

As seen in the graph, both the MLE and MOM estimates are similar and provide a good model for actual death rates.

3.1 Error

MOM and MLE estimates can both work well to estimate parameters for a probability model, but how often are they right and which is better?

To do this, we ran 1,000 simulations using `rbeta()` with parameters $\alpha = 8$ and $\beta = 950$. For each simulation we used MLE and MOM to estimate α and β and graphed a density plot for each estimated parameter.

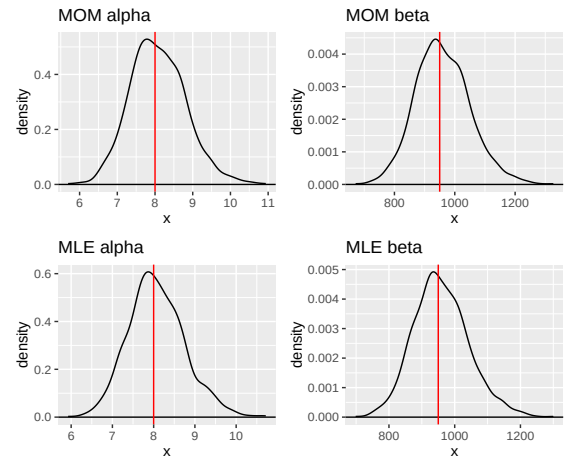


Figure 5: Sampling Distributions of Estimators

Based on the graphs alone, we can tell that the estimators from both methods are accurate most of the time, but there is some variation. Both methods estimators aren't always accurate but it appears that the MLE estimators have less variance. This is further exemplified in the `estimator error table` where we can see that MLE estimators are less biased, and have a lower MSE (Mean Squared Error) than MOM estimators, providing evidence that MLE is the better method of estimating.

References

Erdely, A. and Castillo, I. (2017). `cumstats: Cumulative Descriptive Statistics`. R package version 1.0.
World Bank (2024). *Crude death rate (per 1,000 people)*. Accessed: 2025-04-08.

4 Appendix

	Method	Alpha	Beta	Mean	Variance	Skewness	Kurtosis
1	Formula	2.0	5.0	0.2857	0.0255	0.5963	-0.1200
2	Derived	2.0	5.0	0.2857	0.0255	0.5964	-0.1203
3	Formula	5.0	5.0	0.5000	0.0227	0.0000	-0.4615
4	Derived	5.0	5.0	0.5000	0.0227	0.0000	-0.4618
5	Formula	5.0	2.0	0.7143	0.0255	-0.5963	-0.1200
6	Derived	5.0	2.0	0.7143	0.0255	-0.5964	-0.1203
7	Formula	0.50	0.50	0.5000	0.1250	0.0000	-1.5000
8	Derived	0.50	0.50	0.5000	0.1250	0.0000	-1.5000

Table 1: Comparison of Formula Calculated Values and Derived Values

estimator	mean	bias	precision	MSE
MOM α	8.08	0.08	1.83	0.55
MLE α	8.07	0.07	2.13	0.48
MOM β	960.29	10.29	0.0001	8288.46
MLE β	959.11	9.11	0.0001	7132.70

Table 2: Estimator Error

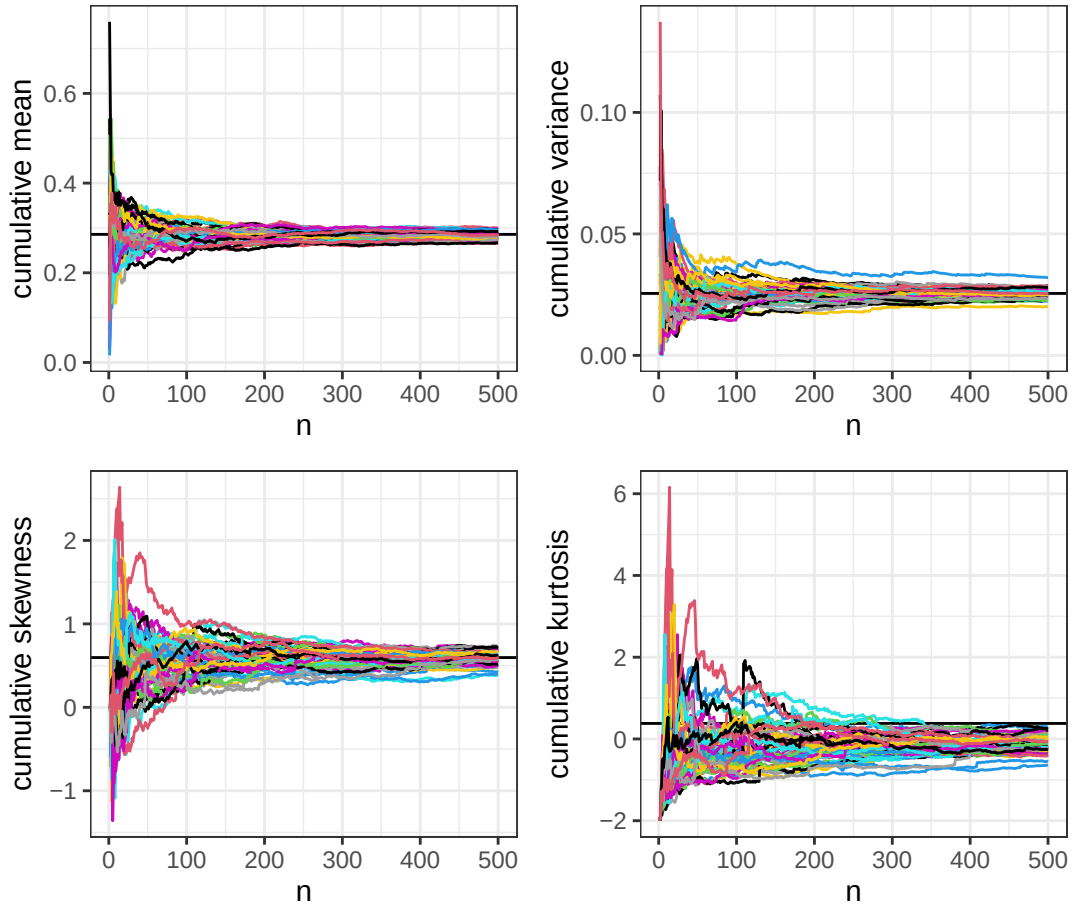


Figure 6: CumStats vs Sample Size